Voice Communications Planning

Quick briefing

Turn the mission into stated voice requirements, choose fixed, tactical or special purpose means for each link, build PACE and recovery into the plan, and chart the equipment, people, funds and maintenance behind it.

Know the facts

Why voice matters · Despite computer networks and satellite communication, voice remains the bedrock of tactical communication (PAM 6-0102 0134). PAM 6-00 (0205) calls the communications system the voice of command.

Sources of requirements · Commander’s guidance or intent, staff guidance (staff are the users), higher, subordinate and adjacent headquarters, SOPs, and the CEOI and CESI (0305). The commander directs the means and the priorities.

CEOI and CESI · The CEOI says what to do; the CESI explains how to do it. Call signs and frequencies come from the CEOI, and how to use them from the CESI.

Signal estimate (METAL) · Mission, Enemy signal capabilities, Troops available, Area of operations, Logistical requirements. Also called the C4S estimate, and revised as the situation changes.

Three categories of communications · Fixed (BONTEX, PANET, AFP FCS, DBTOCS), tactical (HF, VHF and UHF radios, combat net radio with radio over IP) and special purpose (commercial video-conferencing, cellular and satellite phones).

Radio bands · HF/SSB 3 to 30 MHz for long-distance ground communication; VHF/FM 30 to 80 MHz for short distance; UHF/FM 403 to 433 MHz for internal communications.

Resources and maintenance · Chart equipment (0307), personnel (0308) and funds (0309). The SMTF repairs tactical HF and VHF radios up to level 3; SMMART visits brigades, battalions and companies.

Understand the reasoning

A requirement is worth stating only if it supports a decision or an action. Ask which information requirement or task needs the link, and what happens if it fails.

A PACE level is a real fallback only if it fails differently from the level above it. Two radios on the same repeater fail together.

Line of sight drives most voice choices: hills block VHF and UHF, repeaters on high ground extend them, and HF reaches over the horizon with variable conditions.

Philippine Army classroom exercise

In a fictional Philippine Army classroom exercise, a company moves into hills out of VHF line of sight of its command post, and a platoon operates on an island 30 km across open water. The enemy scans clear VHF and has started jamming the battalion VHF net.

Decision · Which means should the planner propose for the two links, and what must the plan add against the enemy?

Worked analysis · For the company in the hills, extend VHF with a repeater on high ground and keep HF as a fallback, because HF reaches over the horizon and does not fail with the repeater. For the island platoon, HF/SSB is the long-distance means, with special purpose means only as gap fillers within COMSEC policy. Against scanning, put encryption-capable radios on sensitive nets and limit what is said on clear nets. Against jamming, rely on electronic protection such as frequency hopping and anti-jamming, and keep a procedure to change frequencies quickly. Record each switch in the PACE table with its trigger.

How to practice

1 · Gather voice requirements from the 0305 sources and state each one with its users, mobility, reach, precedence, security, availability and quality.

2 · Choose the means for each link from the fixed, tactical and special purpose categories, starting from the PAM 6-00 0406 table.

3 · Chart the equipment, personnel and funds available, then rank the requirements that can be supported.

4 · Write a PACE table whose levels fail differently, and add recovery procedures for compromised frequencies, jamming and network attack.

5 · Finalize with reconnaissance, propagation, coverage and gaps, then publish the voice appendix to the C4S annex.

Check yourself

Which facility repairs level 3 HF radios, and which visits companies? · The SMTF repairs tactical HF and VHF radios up to level 3. SMMART visits brigades, battalions and companies.

What must a plan say about compromised frequencies? · It must include procedures to change frequencies quickly, because radios may fall into enemy hands (0311d).

Why must a disaster response net be separate from the operational net? · It is unclassified and unsecured and connects many entities, so it must be different from and separate to the Army’s secured operational net.

Course source

Voice Communications Planning · Student Handout